

Aim: To determine the transition temperature of the given salt hydrate.

Apparatus: Melting point capillary tube, Melting point determination apparatus.

Chemicals Required: Hydrated salts (Sodium sulphate or sodium thiosulphate)

Principle: There are many substances which are capable of existing in more than one crystalline form. These different polymorphous form are not equally stable at a given temperature. However, at a particular temperature, two different forms of substance exist in equilibrium. This temperature is known as the transition temperature of that substance. Not only are transition points associated with polymorphous substances but also with the salt hydrates. When a salt combines with water to form one or more different hydrates, it is found that under a set of given conditions, only one of the hydrates of the salt is stable. Thus when sodium sulphate decahydrate is heated above 33°C , it decomposes into anhydrous sodium sulphate and a saturated solution of this salt. On the other hand, when a saturated salt of sodium sulphate is cooled below 33°C , the anhydrous salt takes up water and forms crystals of decahydrate. Similar relationship is observed with other salt hydrate also.

Procedure:

- (i) Take a small amount of salt hydrate in the melting point capillary
- (ii) Insert the capillary into the melting point apparatus
- (iii) Start heating the sample at very slow rate
- (iv) Record the rise of temperature for every minutes
- (v) Record the temperature of the mixture while cooling at regular interval of time

Observation Table:[illegible]

Plot the temperature versus time graph. Two curves (Fig. 1) with common horizontal portions will be obtained. The horizontal portion corresponds to the transition temperature (T_t) of the given salt hydrate.

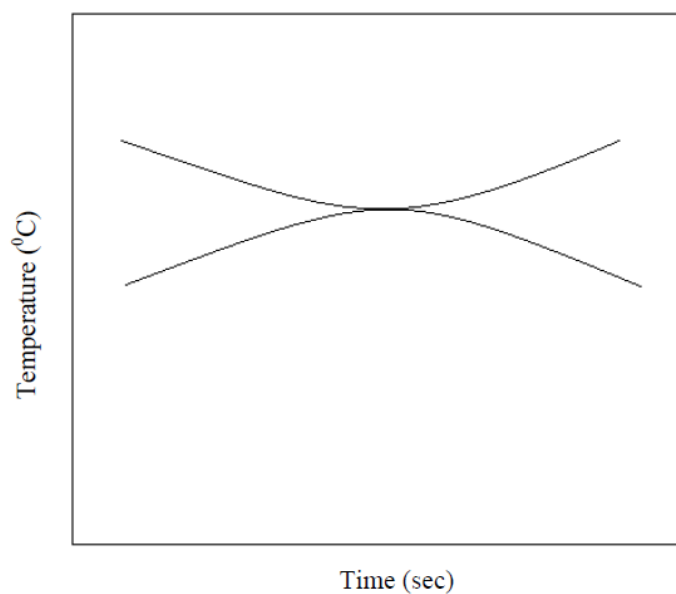


Fig. 1. Temperature-Time graph

Result: The transition temperature of the given substance = $^{\circ}\text{C}$.